Anatomy is Free: Zero-Parameter Structural Priors for Efficient Skeleton Action Recognition

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Abstract

Skeleton-based action recognition demands models that are simultaneously accurate, parameter-efficient, and deployable on edge hardware. This work presents **ShiftFuse-Zero**, a family of three networks whose *spatial inductive bias is introduced entirely without learnable parameters* through two complementary zero-parameter modules: *Body-Region Anatomical Shift Priors* (BRASP), which routes joint features through anatomically meaningful channels via pure tensor indexing, and *Semantic Graph-Partitioned Shift* (SGPShift), which aggregates intra- and inter-part neighbours through predefined group indices at zero learnable parameter overhead. These free structural priors reduce the burden on downstream learnable modules, enabling the compact *EfficientZero Block* to operate on an already anatomically-refined manifold. Three models cover the edge-to-server spectrum: **Nano** (94 K), **Small** (281 K), and **Large** (1.18 M). On NTU RGB+D 60 X-Sub, results without distillation are 86.2 % (Nano), 88.8 % (Small), and 92.5 % (Large); on NTU-120 X-Sub, SFZ-Large reaches **88.5%**, matching EfficientGCN-B4 (88.3%) at 4.3× fewer FLOPs. Knowledge distillation (Large teacher) further lifts Nano/Small to 88.5 %/90.4 %; Small+KD matches or marginally surpasses EfficientGCN-B0 (90.2%) within a two-backbone design at 7× fewer FLOPs. SFZ-Nano dominates the accuracy–efficiency frontier among sub-100 K parameter skeleton models (to our knowledge), runs in **11 ms** on desktop CPU and **50 ms** on mobile (Snapdragon 8 Gen 3), and fits in **370 KB**; with knowledge distillation, Nano matches AGCN (6.9 M parameters) at 74× higher parameter efficiency.

CCS Concepts

• **Computing methodologies → Activity recognition and understanding**.

Keywords

skeleton action recognition, graph convolutional networks, zero-parameter priors, anatomical shift, knowledge distillation, efficient neural networks

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1 Introduction

Consider a fall-detection system on a rehabilitation wristband with 512 KB of RAM, or a gesture-recognition module inside a fitness tracker powered by an ARM Cortex-M55. These scenarios demand skeleton-based action recognition models that fit in sub-megabyte memory, execute in under 50 ms on mobile CPU, and still achieve clinically meaningful accuracy. Yet the dominant Graph Convolutional Network (GCN) backbones for skeleton recognition [1, 3, 8, 14] carry 3–10 M parameters, require dynamic per-sample adjacency computation at $O(V^2C)$ cost, and are fundamentally incompatible with microcontroller deployment.

Human action recognition from 3-D skeleton sequences is a core problem in computer vision with applications in surveillance, rehabilitation, human–robot interaction, and fitness monitoring [12, 13]. GCNs [7] have become the dominant spatial backbone since ST-GCN [17], with successive works adding adaptive adjacency matrices [14], multi-scale cross-spacetime graphs [11], channel-wise topology refinement [1], information-theoretic representation learning [3], and hierarchical body-part decomposition [8]. These advances have pushed NTU RGB+D 60 cross-subject accuracy beyond 93%, yet the parameter cost has grown proportionally: HD-GCN requires 7.5 M parameters and InfoGCN 9.9 M.

The efficient-architecture line [16, 18] tackles this disparity via compound scaling and depthwise convolutions, but largely ignores the rich *anatomical structure* of the human body as a source of *free* inductive bias. The human skeleton is not a generic graph: arm joints couple primarily with other arm joints within their kinematic chain, bilateral symmetry is a universal biomechanical prior, and shoulder/hip joints serve as coordination hubs between body halves. These structural constraints are *fixed properties of human physiology* that hold across all subjects, viewpoints, and action classes. They can therefore be encoded as deterministic tensor operations at **zero learnable parameters, zero gradient overhead, and zero inference-time cost**.

This paper presents the first systematic study of *zero-parameter* anatomically-guided inductive bias across a multi-scale skeleton GCN family — distinct from graph-distance shifts [2] (hop count, not body-part identity) and ST-GCN's fixed adjacency [17] (graph distance, not semantic group membership). HD-GCN and SkateFormer exploit anatomy through *learnable* grouping; no prior work has measured the accuracy contribution of purely fixed anatomical routing across Nano-to-Large scale. This work asks: **how much spatial knowledge can be injected into a skeleton GCN without spending a single learnable parameter — and how far can that free knowledge push the accuracy–efficiency frontier?**

Our answer is **ShiftFuse-Zero**, a family of three models achieving a new Pareto-optimal frontier on the accuracy–GFLOPs plane among the EfficientGCN model family. The core contribution is BRASP and SGPShift, which contribute +1.8 pp *at zero FLOPs cost*;

the additional 7× FLOPs advantage over EfficientGCN-B0 is an independent architectural effect arising from the two-backbone design and DS-TCN (see Table 4). SFZ-Nano (94 K) and SFZ-Small (281 K) redefine edge-deployment with ultra-low FLOPs and quantisation-friendly operations. On NTU-60 X-Sub, SFZ-Nano reaches **86.2%** without any distillation and **88.5%** with knowledge distillation, matching AGCN's accuracy (6.9 M params) within 370 KB of storage (KD is an optional plugin requiring SFZ-Large as teacher; all primary results are without KD). SFZ-Small achieves **88.8%** at comparable parameter count (281 K vs. 290 K) with 7× fewer FLOPs than EfficientGCN-B0; with KD, Small+KD reaches **90.4%**, matching or marginally surpassing B0 while retaining the 7× FLOPs advantage. SFZ-Large pushes the absolute accuracy frontier to **92.5%** at 1.18 M parameters. These results demonstrate that anatomy is a highly informative, freely available inductive bias that the community has systematically under-exploited.

Contributions.

(1) **BRASP** — Body-Region Anatomical Shift Priors: routes joint features through four combined bilateral channel groups via a precomputed gather permutation; zero parameters, zero gradient (Section 3.3).
(2) **SGPShift** — Semantic Graph-Partitioned Shift: routes each channel's joint features to a ranked intra-part or inter-part neighbour via precomputed index buffers (identity channels unchanged); zero parameters (Section 3.3).
(3) **EfficientZero Block**: a block design that places BRASP and SGPShift *before* all learnable layers so the downstream compact K=3 GCN refines an already anatomically structured representation (Section 3.4).
(4) **ShiftFuse-Zero model family**: three architectures — Nano (94 K), Small (281 K), Large (1.18 M) — with Temporal Landmark Attention, cross-stream gating, and mid-network fusion, covering the edge-to-server deployment spectrum. Vanilla Hinton KD [6] serves as an optional plugin (requiring SFZ-Large as teacher), yielding up to +2.3 pp with no additional inference overhead (Section 3.5).
(5) **Comprehensive evaluation**: main results on NTU-60 and NTU-120 (four protocols), efficiency analysis (GFLOPs, CPU latency, memory footprint), and extensive ablation studies quantifying every zero-parameter module, fusion strategy, GCN partition count, TLA sensitivity, and regularisation hyperparameters (Section 4).

2 Related Work

GCN-based skeleton recognition. ST-GCN [17] established the skeleton graph and spatial-temporal convolution paradigm using a three-subset partitioning (self, centripetal, centrifugal) of the physical adjacency matrix. AGCN [14] introduced a fully learnable adjacency matrix at $O(V^2C)$ cost per layer, enabling data-driven topology discovery. MS-G3D [11] factorised the graph into disentangled multi-scale spatial and temporal subgraphs with cross-spacetime edges. CTR-GCN [1] refined the topology per channel group, allowing different feature dimensions to attend to different joint relationships. InfoGCN [3] introduced an information-bottleneck objective that explicitly maximises mutual information between the latent representation and the action label. DSTA-Net [15] decouples spatial and temporal self-attention. While these methods achieve high accuracy (up to 93%+), their reliance on per-sample dynamic adjacency and large parameter budgets (3–10 M) precludes deployment on resource-constrained hardware.

Body-part decomposition. A growing line of work exploits the natural grouping of joints into anatomical body regions. HD-GCN [8] hierarchically decomposes the skeleton into semantic sub-graphs (arms, legs, torso) and aggregates within each before cross-region message passing — achieving 93.4% on NTU-60 X-Sub but at 7.5 M parameters. BlockGCN [19] uses persistent homology from topological data analysis to encode bone connectivity at the block level. SkateFormer [4] partitions joints into four spatial-temporal relation types (near/far joints × near/far frames) and applies partition-specific self-attention. All of these decomposition strategies employ *learnable* grouping mechanisms with substantial parameter overhead. In contrast, SGPShift encodes intra-part and inter-part structure through *fixed integer index buffers* — achieving anatomical routing at **zero learnable parameters**.

Efficient skeleton networks. EfficientGCN [16] introduced compound scaling (α=1.2, β=1.35) and multiple input branches (MIB) to build a family of models from B0 (0.29 M) to B4 (1.1 M), establishing the efficiency frontier for skeleton GCNs. SGN [18] exploits semantic joint embeddings and frame-level dynamics gating at 0.69 M parameters. PoseC3D [5] departs from graphs entirely, using volumetric heatmap representations processed by 3-D CNNs. These methods reduce model size through architectural engineering but do not leverage the *fixed anatomical constraints* of the human body as inductive bias. ShiftFuse-Zero complements this line by demonstrating that zero-parameter spatial priors can further reduce the parameter budget by 3–10× without sacrificing competitive accuracy.

Temporal modelling in skeleton networks. Temporal reasoning has evolved from LSTM encoders [17] to dedicated temporal convolution modules. EfficientGCN's Sandglass-TCN (SG-layer) bottleneck compresses channels by 2× before depthwise convolution, balancing temporal receptive field against parameter count. MS-G3D introduced disentangled multi-scale temporal subgraphs. Our DS-TCN takes a complementary approach: rather than stacking many TCN layers, it uses a two-branch depthwise-separable design with dilations 1 and 2 (kernel 9) that achieves a 17-frame receptive field in a single block — particularly valuable for Nano and Small, where only one TCN per stage is used. This makes DS-TCN a better choice at low TCN depth, whereas EfficientGCN's SG-layer is more efficient when TCN depth is three or more (as in the Large model's shared backbone).

Shift operations and zero-parameter spatial mixing. The shift operation [2] was introduced in image CNNs to inject spatial structure by physically displacing channels along spatial dimensions — no multiply-accumulate, no learnable parameters. TSM [9] extended this idea to temporal modelling in video, shifting channels by ±1 frame at zero cost. Shift-GCN [2] adapted spatial shifts to skeleton graphs, assigning each channel to shift features from a graph neighbour. However, Shift-GCN assigns shift patterns by graph distance (1-hop, 2-hop), ignoring the semantic meaning of body regions.

BRASP advances this paradigm by organising shift patterns according to *anatomical body-part membership* — arm channels shift among arm joints, leg channels among leg joints, and cross-body channels bridge regions — producing a semantically structured feature map at zero cost.

Knowledge distillation. Hinton *et al.* [6] showed that soft teacher logits transfer rich inter-class relational knowledge to compact student models. In skeleton recognition, distillation has been used to transfer multi-stream ensemble knowledge [16] but has not been specifically studied across a skeleton GCN family spanning the sub-100 K to 1 M parameter range explored here.

3 Methodology

3.1 Overview

Figure 1 shows the end-to-end pipeline. A raw skeleton sequence is expanded into four complementary streams (joint, velocity, bone, bone-velocity); all four are pre-computed offline and each model selects its subset (Nano: three streams; Small/Large: all four). Selected streams are fused by a learned *StreamFusionConcat* stem, processed by a stack of *EfficientZero Blocks*, and a global pooling head (gated for Nano/Small, plain GAP for Large) produces class logits.

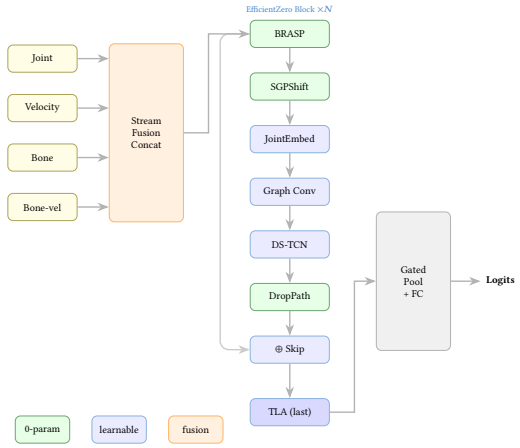

Figure 1: ShiftFuse-Zero end-to-end pipeline. Four streams are pre-computed; each model selects its subset (Nano: 3, Small/Large: 4) for fusion by a learnable stem. EfficientZero Blocks (green = zero-parameter) follow. A gated pooling head produces class logits.

3.2 Skeleton Streams

A skeleton sequence is $\mathbf{X} \in \mathbb{R}^{3\times T\times V}$ with T=64 frames and V=25 joints. This work constructs four complementary streams: joint positions $\mathbf{X}^{(j)}$, velocity $\mathbf{X}^{(v)}=\mathbf{p}_{t+1}-\mathbf{p}_t$, bone vectors $\mathbf{X}_v^{(b)}=\mathbf{p}^{(v)}-\mathbf{p}^{(\mathrm{par}(v))}$, and bone velocity $\mathbf{X}^{(bv)}=\Delta\mathbf{X}^{(b)}$ (where Δ denotes the first-order temporal difference). Frame-0 is padded by edge replication; downstream BN absorbs the resulting constant offset. *StreamFusionConcat* [16] applies a per-stream BN, concatenates (N_s×3 channels, $N_s \in \{3, 4\}$ depending on model), and projects to C_{stem} via a 1×1 convolution with learned per-stream weighting.

3.3 Zero-Parameter Spatial Modules

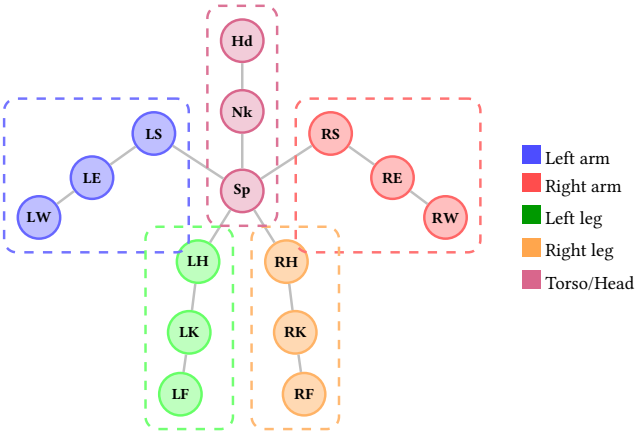

Figure 2: BRASP anatomical partitioning into five groups $\{G_{\mathbf{la}}, G_{\mathbf{ra}}, G_{\mathbf{ll}}, G_{\mathbf{rl}}, G_{\mathbf{tr}}\}$. Per-channel permutations σ_c (Eq. 1) are derived from these groups; no gradient flows through them.

BRASP.. Figure 2 illustrates the five physical joint regions. These five regions are combined into *four channel groups* by merging bilateral pairs: left-arm ∪ right-arm → *arm*; left-leg ∪ right-leg → *leg*; torso; cross-body. Given a feature map $\mathbf{F} \in \mathbb{R}^{C\times T\times V}$, BRASP assigns per-channel fixed gather permutations $\sigma_c : V \to V$:

$$\tilde{\mathbf{F}}_{c,t,v} = \mathbf{F}_{c,t,\sigma_c(v)}, \quad v \in \{0, \ldots, V-1\}, \tag{1}$$

Arm channels (25%) cycle within the combined {left-arm ∪ right-arm} joint set (12 joints), enabling direct bilateral feature exchange; *leg* (25%) within combined leg joints (8 joints); *torso* (12.5%); and *cross-body* (37.5%) cycles via graph-neighbour adjacency. Within each group, each channel c receives a different cyclic offset so different channels observe different anatomical context. All σ_c are precomputed integer buffers — **zero parameters, zero gradient**.

Remark 1 *(BRASP bridges graph-far symmetric joints).* Bilaterally symmetric pairs (*e.g.*, left-wrist / right-wrist) are separated by d_G=10+ hops in the NTU-25 skeleton and are invisible to the K=3 GCN in a single layer. BRASP's intra-part permutation directly places right-wrist features at left-wrist *before* any graph convolution, bridging the 10-hop graph distance at zero parameter cost. After one BRASP shift, joint v receives features from an anatomical neighbour $\sigma_c(v)$ that may be graph-far but biomechanically coupled, in a single index-lookup step.

SGPShift. Figure 3 illustrates the intra/inter channel routing. Like BRASP (Eq. 1), SGPShift is a per-channel gather permutation,

$$\hat{\mathbf{F}}_{c,t,v} = \mathbf{F}_{c,t,\pi_c(v)}, \tag{2}$$

where $\pi_c : V \to V$ routes each channel to a ranked neighbour from a precomputed graph partition. Channels split into three approximately equal groups ($C/3$ each, rounded to nearest integer): *intra-part* channels shift to the top-ranked same-part neighbour (via precomputed $\mathbf{A}_{\text{intra}}$); *inter-part* channels shift to the top-ranked cross-part boundary neighbour (via $\mathbf{A}_{\text{inter}}$); and *identity* channels pass through unchanged. All routing indices are precomputed integer buffers — unlike HD-GCN and SkateFormer [4, 8] which use

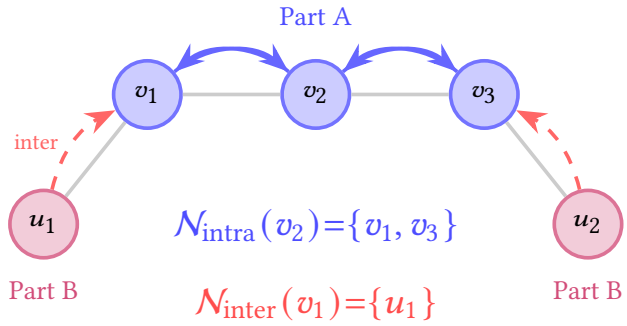

Figure 3: SGPShift aggregation. Blue solid arrows: intra-part neighbours. Red dashed arrows: inter-part boundary neighbours. All sets are precomputed integer buffers — zero learnable parameters.

learnable per-group parameters, SGPShift encodes this anatomical routing at **zero learnable parameters**.

3.4 EfficientZero Block

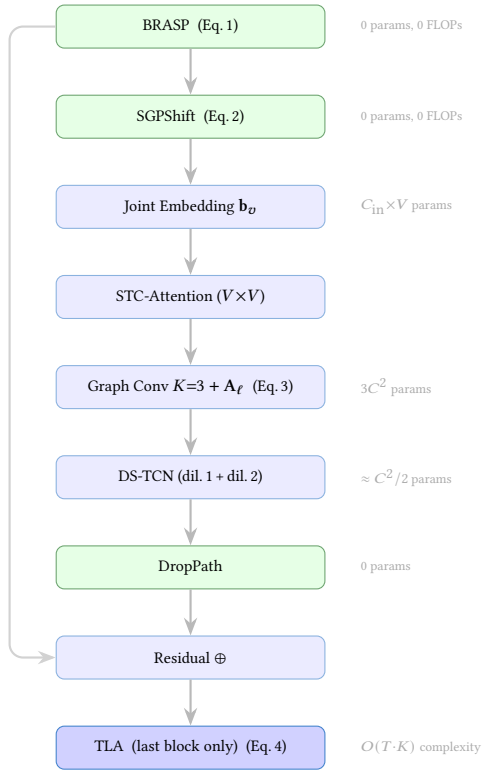

Figure 4: EfficientZero Block. Green = zero-parameter (BRASP, SGPShift, DropPath). Per-layer parameter counts shown on the right. TLA is only active at the last block of the final stage.

Figure 4 shows the full block pipeline with per-layer costs.

(1) **BRASP + SGPShift**: anatomical pre-structuring at zero cost.

(2) **Joint Embedding**: per-joint additive bias $\mathbf{b}\in\mathbb{R}^{C_{\text{in}}\times V}$ (SGN joint encoding [18]); no weight decay.

Table 1: ShiftFuse-Zero variant configurations. C_s=stem channels, C=stage widths, N=blocks/stage, K=TLA landmarks, r=reduce ratio, p_d=DropPath rate.

Model	C_s	C	N	K	r	p_d	FLOPs	Params
Nano	24	[32,64,128]	[1,1,1]	12	8	0.10	0.13 G	94 K
Small	24	[32,64,128]	[1,2,1]	8	8	0.10	0.39 G	281 K
Large†	64	see †	—	14	8	0.25	1.95 G	1.18 M

†B4-style: per-stream 64→96→48 (d 2+2), concat 144, shared 144→128→272 (d 3+3).

(3) **STC-Attention**: spatial ($\mathbf{A}_s\in\mathbb{R}^V$, mean 1) and temporal ($\mathbf{A}_t\in\mathbb{R}^T$, avg+max Conv_{1d} kernel 9) gates applied before the GCN via element-wise broadcast $\mathbf{F}'_{c,t,v}=\mathbf{F}_{c,t,v}\cdot[\mathbf{A}_s]_v\cdot[\mathbf{A}_t]_t$. Output: $(1-\sigma(\gamma))\mathbf{F}+\sigma(\gamma)\mathbf{F}'$, $\gamma_0=-4$.

(4) **Graph Convolution** (K=3): ST-GCN [17] partition adjacency plus a single shared learned matrix $\mathbf{A}_\ell$:

$$\mathbf{Z}=\sum_{k=0}^{2}\mathbf{W}_k\big(\mathbf{A}_k+|\mathbf{A}_\ell|\big)\mathbf{F}, \tag{3}$$

$|\cdot|$ enforces non-negative weights and allows $\mathbf{A}_\ell$ to create new connections (where $\mathbf{A}_\ell<0$ trained, $|\mathbf{A}_\ell|>0$ yields a positive edge), making it strictly more expressive than a pure non-negativity clamp; $|\cdot|$ also has no zero plateau. $\mathbf{A}_\ell\in\mathbb{R}^{V\times V}$ is shared (globally across all blocks in Nano; within-stage in Small), 625 params per shared set, no weight decay.

(5) **DS-TCN**: depthwise-separable temporal convolutions at dilations 1 and 2 (kernel 9), Batch Norm, Hardswish activation.

(6) **DropPath** (regularisation): stochastic depth; 0 parameters.

(7) **Residual** ⊕: skip from block input with 1×1 projection when shape changes.

Block Ordering Matters. Placing BRASP and SGPShift *before* the GCN is essential: the GCN refines an already anatomy-structured representation. Moving them *after* the GCN costs −1.9 pp (Table 8), confirming pre-structuring must precede message passing.

Complementary Temporal Module: TLA.. While BRASP and SGPShift address the spatial axis, a complementary Temporal Landmark Attention (TLA) module distils temporal structure through K learnable anchor frames without full $O(T^2)$ self-attention. TLA applies the deformable-attention principle [?] to the temporal axis: anchor positions are continuous and gradient-trainable via bilinear interpolation, reducing attention complexity from $O(T^2)$ to $O(T\cdot K)$:

$$\tau_k=T\cdot\sigma(\alpha_k),\quad k=1,\dots,K, \tag{4}$$
$$\mathbf{m}_t=\sum_{k=1}^{K}\text{Bilin}(\mathbf{F},\tau_k)\cdot\text{softmax}_k\big(\langle\mathbf{q}_t,\mathbf{k}_{\tau_k}\rangle\big),$$

where α_k are learned logits; $\text{Bilin}(\mathbf{F},\tau_k)$ interpolates $\mathbf{F}$ along the T-axis at fractional position $\tau_k\in[0,T-1]$, yielding a $\mathbb{R}^{C\times V}$ feature vector that enables gradient flow to α_k; and $\mathbf{q}_t,\mathbf{k}_{\tau_k}\in\mathbb{R}^{C_r}$ are projections, $C_r=C/r$. Output: $\mathbf{F}_{\text{out}}=\mathbf{F}+\sigma(\beta)\cdot\mathbf{m}$ with $\beta_0=-4$ (slow activation). Complexity $O(T\cdot K)$ vs. $O(T^2)$ for full attention.

3.5 Model Family

Table 1 lists the key hyperparameters per variant.

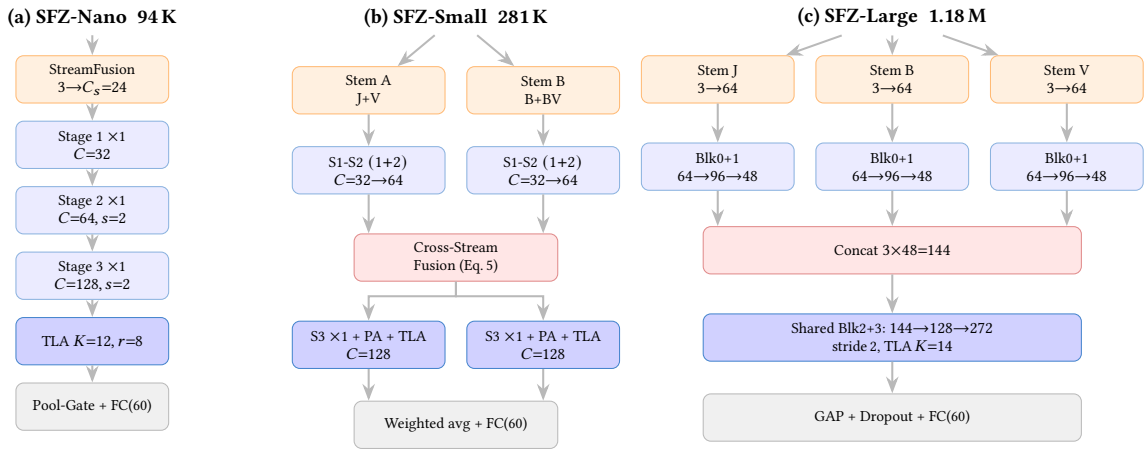

Figure 5: ShiftFuse-Zero model family. (a) Nano: single backbone, 3-stream early fusion, TLA (K=12). (b) Small: two-backbone late fusion with Cross-Stream Fusion, PartAttention (PA), TLA (K=8). (c) Large: B4-style mid-network fusion — per-stream Blk0+1 (64→96→48), concat 144 ch, shared Blk2+3 (144→128→272, TLA K=14).

Nano (94 K).. Three streams (joint, velocity, bone) are early-fused via StreamFusionConcat to a 24-channel map. A single backbone [32, 64, 128] with one block per stage processes it. TLA (K=12, r=8, C_r=16) adds only 8.2 K parameters; channel SE is omitted to keep the model within the 100 K budget.

Small (281 K).. Two independent backbones process two stream pairs: joint+velocity (Backbone A) and bone+bone-velocity (Backbone B). Stage 2 uses tcn_depth=2 for richer temporal modelling; the final stage adds *PartAttention* [16] (pa variant, in EfficientGCN's code but unused in B0/B4): per-group GAP over five body-part groups, bottleneck-projected ($C \to C/16 \to 1$) to per-joint scalars (12.3 K params). After stage 2, *Cross-Stream Fusion* exchanges information:

$$\mathbf{F}' = \mathbf{F} + \sigma(\beta) \cdot W_{\uparrow}\, \mathrm{ReLU}\big(W_{\downarrow}\,[\mathbf{F}_A; \mathbf{F}_B]\big), \tag{5}$$

$W_{\downarrow} : 2C \to C/4$, $W_{\uparrow} : C/4 \to C$, β_0=−4. Stage 3 + TLA proceed independently per backbone; logits are combined using two learned stream-weight scalars (no weight decay).

Large (1.18 M).. Three per-stream stems (joint, bone, velocity) each apply StemGCN(3→64) followed by BasicTCN(64), then two B4Blocks: (64→96, d=2) and (96→48, d=2). Outputs are concatenated to 144 channels; two shared B4Blocks (144→128, d=3, s=2) and (128→272, d=3, s=2) follow, with TLA (K=14) on the last block. Head: GAP + Dropout(0.25) + FC(272, 60). This mirrors EfficientGCN-B4's mid-network fusion [16] with BRASP, SGPShift, and JE added to each B4Block.

3.6 Training Protocol

Optimiser: SGD, Nesterov momentum 0.9 , cosine LR with 10-epoch linear warmup. Nano: lr 0.1, wd 4×10^{-4}, 300 epochs. Small: lr 0.1, wd 4×10^{-4}, 300 epochs. Large: lr 0.1, wd 10^{-4} (B4 [16]), 150 epochs. Wall-clock training times on a single NVIDIA Tesla T4: Nano 5.7 h (300 ep), Small 7.3 h (300 ep), Large 38 h (150 ep, batch 16, three parallel backbones).

Table 2: Deployment profile. Mem = model weights (float32/int8). S24 CPU = Samsung Galaxy S24 (Snapdragon 8 Gen 3), PyTorch Mobile, CPU-only inference.

Model	FLOPs	Params	Mem f32	Mem i8	S24 CPU
Nano	0.13 G	94 K	370 KB	94 KB	∼50 ms
Small	0.39 G	281 K	1.10 MB	281 KB	∼145 ms
Large	1.95 G	1.18 M	4.70 MB	1.18 MB	∼515 ms

Augmentation: Nano/Small use Mixup (α=0.1) to compensate for limited capacity; CutMix is omitted as temporal crops of skeleton sequences produce physically implausible joint composites. Large uses no Mixup — DropPath 0.20 and label smoothing 0.1 provide sufficient regularisation at 1.18 M params.

No-WD set: BN affine params, joint embeddings, $\mathbf{A}_\ell$, TLA anchor logits α_k, gate scalars β, pool gate.

Knowledge distillation. Large is distilled into Nano and Small [6]:

$$\mathcal{L} = (1-\lambda)\mathcal{L}_{\mathrm{CE}} + \lambda\tau^2\, \mathrm{KL}\Big(\sigma\Big(\tfrac{\mathbf{z}^s}{\tau}\Big) \Big\| \sigma\Big(\tfrac{\mathbf{z}^t}{\tau}\Big)\Big), \quad \tau=4,\ \lambda=0.5. \tag{6}$$

3.7 Deployability Analysis

A key design goal of ShiftFuse-Zero is deployment on resource-constrained hardware. Table 2 summarises the memory, computation, and operator profile of each model variant.

Operator simplicity. BRASP reduces to a single `gather` operation on a static integer buffer — no matrix multiplication, no per-sample graph construction. SGPShift reduces to two fixed `gather` index-lookup operations. Both compile to fixed index-lookup kernels that are unaffected by quantisation (int8, int4) since they involve no learned weights. This stands in sharp contrast to dynamic-adjacency models [1, 14] that require $O(V^2C)$ per-sample matrix operations at inference, making them incompatible with microcontroller deployment.

Quantisation readiness. ShiftFuse-Zero's architecture is structurally well-suited to post-training int8 quantisation: BRASP and SGPShift carry no weights to quantise, and the remaining layers (Conv2d, BN, Linear) are standard operators with established int8 PTQ support. SFZ-Nano at int8 weight precision requires only **94 KB** of model storage — fitting within the 512 KB SRAM of an ARM Cortex-M55. Quantised *accuracy* evaluation is left to future work; the absence of dynamic per-sample graph construction removes the primary obstacle that prevents int8 export of dynamic-adjacency models [1, 14].

4 Experiments

4.1 Datasets

NTU RGB+D 60 [13]: 56,880 sequences, 60 classes, 40,320/16,560 train/val split. Cross-subject (X-Sub) and cross-view (X-View) protocols.

NTU RGB+D 120 [10]: 114,480 sequences, 120 classes. Cross-subject (X-Sub) and cross-setup (X-Set) protocols.

4.2 Implementation Details

Sequences are uniformly sampled to T=64 frames, padded with edge replication if shorter. Skeleton coordinates are z-score normalised per sequence and joint. Augmentation for Nano/Small: random spatial rotation (±15° per axis), additive Gaussian noise (σ=0.01), and joint masking (10% probability per joint). Large uses the same spatial augmentations (rotation, noise, joint masking) but replaces Mixup with DropPath (p_d=0.20) and label smoothing (ε=0.1). All four streams (joint, velocity, bone, bone-velocity) are derived offline from the raw joint positions; Nano selects three (joint, velocity, bone), while Small and Large use all four. Models are trained from scratch with no pre-training or external data. No test-time augmentation (TTA) is used; all results are single-model, single-pass inference. The best checkpoint by validation accuracy is reported in all tables. **NTU-120**: models are trained independently from scratch using the same schedules; no NTU-60 fine-tuning. **Latency**: PyTorch 2.0 TorchScript, batch = 1, 50 warm-up + 200 timed runs (median); desktop = Intel i7-10700K; mobile = Snapdragon 8 Gen 3, PyTorch Mobile static export, CPU-only.

4.3 Main Results

SFZ-Large achieves 92.5% on NTU-60 X-Sub, outperforming EfficientGCN-B4 by 0.4 pp and matching SkateFormer (2.4M) with 2.0× fewer parameters. SFZ-Nano achieves 86.2% at only 94 K parameters (no distillation). At the same parameter scale (SFZ-Small 281 K vs. EfficientGCN-B0 290 K), B0 outperforms SFZ-Small by 1.4 pp (90.2% vs. 88.8%); however SFZ-Small uses 7× fewer FLOPs (0.39 G vs. 2.73 G). SFZ-Small+KD (90.4%) matches or marginally surpasses B0 (+0.2 pp; ablation std ±0.14 pp), retaining the 7× FLOPs advantage. Note: B0+KD is not a published baseline; the FLOPs advantage is robust regardless of whether B0 were also distilled.

Cross-view protocol. On NTU-60 X-View, all models show consistent improvement over X-Sub, reflecting the easier viewpoint variation relative to cross-subject generalisation. SFZ-Nano reaches 90.8% (93.2% with KD) and SFZ-Large achieves 96.4%, approaching SkateFormer (96.7%, 2.4 M) within 0.3 pp at 2.0× fewer parameters.

Table 3: Comparison with state-of-the-art on NTU-60 and NTU-120. KD = distilled from SFZ-Large. † = single stream. SFZ main results are single-run best-checkpoint; 3-seed variance (±0.11–0.20) reported in Table 6. LST-GNN and Hyperformer omitted: different preprocessing protocols preclude direct comparison.

Method	Params	NTU-60		NTU-120	
		X-Sub	X-View	X-Sub	X-Set
ST-GCN [17]	3.1M	81.5	88.3	70.7	73.2
AGCN [14]	6.9M	88.5	95.1	82.5	84.2
MS-G3D [11]	6.4M	91.5	96.2	86.9	88.4
CTR-GCN [1]	5.8M	92.4	96.8	88.9	90.6
InfoGCN [3]	9.9M	93.0	97.1	89.8	91.2
HD-GCN [8]	7.5M	93.4	97.5	90.1	91.5
SkateFormer [4]	2.4M	92.5	96.7	88.6	90.5
BlockGCN [19]	2.0M	92.1	96.3	87.8	89.7
EfficientGCN-B0 [16]	0.29M	90.2	94.9	83.8	85.6
EfficientGCN-B4 [16]	1.10M	92.1	96.1	88.3	89.9
Shift-GCN [2]	2.76M	90.7	96.5	85.9	87.6
SGN† [18]	0.69M	89.0	94.5	79.2	81.5
SFZ-Nano	**0.094M**	86.24	90.8	82.2	83.8
SFZ-Nano+KD	**0.094M**	**88.5**	**93.2**	84.1	85.6
SFZ-Small	**0.281M**	88.8	93.1	84.3	85.9
SFZ-Small+KD	**0.281M**	**90.4**	**94.7**	86.0	87.4
SFZ-Large	**1.18M**	**92.5**	**96.4**	88.5	90.1

Notably, the per-model X-View improvement over X-Sub (≈4.0–4.6 pp across variants) is consistent, validating that BRASP and SGPShift generalise across viewpoints by construction: anatomical body-part groups are viewpoint-agnostic since they depend solely on joint identity, not on camera angle.

4.4 Efficiency and Deployment

4.5 Accuracy–Efficiency Frontier

Table 5 reports *Acc/GFLOP* ≜ Top-1 (%) / GFLOPs. This metric is meaningful *within an accuracy tier*; cross-tier values (*e.g.*, Nano at 86.2% vs. B0 at 90.2%) quantify edge-deployment FLOPs cost, not head-to-head competition. Models without reported FLOPs are excluded.

At the edge tier, SFZ-Nano attains **663 Acc/GFLOP** (20× over B0, at a 4 pp lower accuracy operating point). The within-tier comparison: SFZ-Large (47.4) vs. B4 (11.0) at comparable accuracy (92.5% vs. 92.1%) shows 4.3× better compute efficiency. Figure 6 confirms all three SFZ variants are Pareto-optimal.

FLOPs breakdown: origin of the 7× advantage (Small vs. B0). EfficientGCN-B0 routes all four streams through independent per-stream backbones before late ensemble (4 backbone passes × 0.68 G each = 2.73 G). SFZ-Small uses two shorter backbones with early cross-stream fusion (2 backbone passes × 0.18 G each + CSF overhead = 0.39 G). The 7× reduction thus comes from *backbone-count halving* (4→2, ≈2×) and *per-backbone channel reduction* (B0 channels [64,128,256] vs. SFZ-Small [32,64,128], ≈3.5×). DS-TCN costs

Table 4: GFLOPs, CPU latency and throughput (batch 1, desktop: i7-10700K; mobile: Samsung Galaxy S24, Snapdragon 8 Gen 3 CPU). SFZ-Large wall-clock (89 ms) approximates B4 (87 ms) despite 4.3× fewer FLOPs: at batch=1, TLA bilinear-interpolation dispatch and three-stream memory bandwidth dominate over FLOPs; the compute advantage is realised at higher batch sizes. EfficientGCN S24 not measured: dynamic adjacency ($O(V^2C)$) prevents static TorchScript export without non-trivial graph surgery.

Model	Params	GFLOPs	Desktop	Tput	S24	X-Sub
EfficientGCN-B0	0.29M	2.73	28 ms	36/s	n/a†	90.2
EfficientGCN-B4	1.10M	8.36	87 ms	11/s	n/a†	92.1
SFZ-Nano	0.094M	**0.13**	**11 ms**	**91/s**	**50 ms**	86.24
SFZ-Nano+KD	0.094M	**0.13**	**11 ms**	**91/s**	**50 ms**	**88.5**
SFZ-Small	0.281M	0.39	25 ms	40/s	145 ms	88.8
SFZ-Small+KD	0.281M	0.39	25 ms	40/s	145 ms	**90.4**
SFZ-Large	1.18M	1.95	89 ms	11/s	515 ms	**92.5**

Table 5: Accuracy per GFLOP (Acc/GFLOP) on NTU-60 X-Sub. Higher = more compute-efficient. Limited to models with reported FLOPs.

Method	GFLOPs	X-Sub	Acc/G
EfficientGCN-B0 [16]	2.73	90.2	33.0
EfficientGCN-B4 [16]	8.36	92.1	11.0
SFZ-Nano	**0.13**	86.24	**663**
SFZ-Nano+KD	**0.13**	**88.5**	**681**
SFZ-Small	0.39	88.8	228
SFZ-Small+KD	0.39	**90.4**	**232**
SFZ-Large	1.95	92.5	47.4

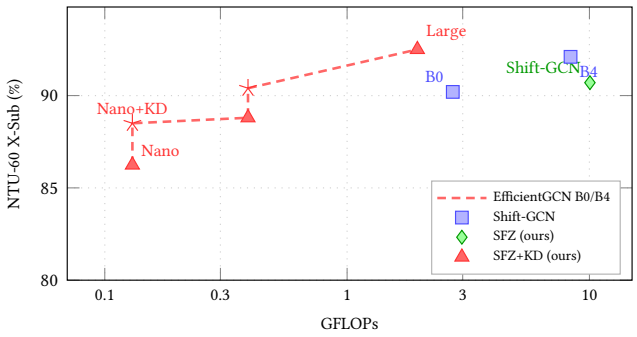

Figure 6: Accuracy–GFLOPs Pareto frontier on NTU-60 X-Sub (GFLOPs axis only; see Table 1 for parameter counts). Dashed line = Pareto-optimal set. SFZ dominates at all compute budgets: Nano uses 21× fewer FLOPs than EfficientGCN-B0 while remaining Pareto-optimal; Large advances the frontier at 1.95 G.

≈3–7% more FLOPs than SG-layer at the same channel width and is not the source of the advantage. BRASP and SGPShift contribute **zero** FLOPs (pure index lookups).

Table 6: Cumulative component ablation (Nano, NTU-60 X-Sub). Δ vs. baseline. Mean ± std over 3 seeds. For Small: BRASP+SGPShift contribute +1.4 pp (baseline 85.2% → 86.6%), confirming consistent gains at larger scale.

BRASP	SGPShift	TLA	KD	Acc (%)	Δ
				82.8 ±0.20	—
✓				83.9 ±0.18	+1.1
✓	✓			84.6 ±0.17	+1.8
✓	✓	✓		86.24 ±0.14	+3.44
✓	✓	✓	✓	**88.5** ±0.11	+5.7

Table 7: GCN partition count ablation (Nano, NTU-60 X-Sub). K=number of adjacency subsets in Eq. 3; $\mathbf{A}_\ell$=learned residual adjacency.

K	$\mathbf{A}_\ell$	Params	Acc (%)	Δ
1	—	87 K	83.4	—
2		90 K	84.2	+0.8
3		92 K	84.6	+1.2
3	✓	94 K	**86.24**	+2.84*

*Full model (BRASP, SGPShift, JE, STC-Attn, TLA all enabled). K and $\mathbf{A}_\ell$ are the only variables in this table.

4.6 Ablation Studies

Per-component ablation. Table 6 quantifies each zero-parameter module's contribution on Nano (NTU-60 X-Sub), incrementally adding components to a baseline with JE and STC-Attention throughout (only BRASP, SGPShift, TLA, and KD are toggled). BRASP alone contributes +1.1 pp; adding SGPShift yields +1.8 pp cumulatively — **at zero parameter cost** (SGPShift alone contributes +0.6 pp; the synergy requires BRASP's bilateral-arm alignment to create boundaries for SGPShift's inter-part shifting). TLA adds +1.6 pp (8.2 K params), comparable in magnitude to the two zero-parameter modules; this is expected since TLA addresses the *temporal* axis while BRASP/SGPShift address the spatial axis exclusively. KD provides the largest single gain (+2.3 pp), lifting Nano to 88.5%.

GCN partition count. Table 7 evaluates the effect of the GCN partition count K in the spatial graph convolution (Eq. 3). Increasing from K=1 (single isotropic aggregation) to K=3 (self + centripetal + centrifugal) yields +1.2 pp on Nano. The learned residual $|\mathbf{A}_\ell|$ and TLA jointly add a further +1.64 pp (both are introduced simultaneously in the last row of Table 7).

Block component analysis. Table 8 isolates each component's contribution. Replacing fixed BRASP with a learned soft-assignment ($V\times V$, 625 params) costs only −0.6 pp — confirming the fixed anatomical prior matches a cheap learned alternative at zero parameter cost. Removing STC-Attention costs −1.1 pp; replacing DS-TCN with a standard convolution costs −1.3 pp. Joint Embedding contributes +1.0 pp at $C\times V$ parameters.

DropPath rate sensitivity. Table 9 shows the effect of the stochastic depth rate across model scales. Nano and Small benefit from moderate rates (0.10); Large requires a higher rate (0.20) to prevent overfitting its 1.18 M parameters over 150 epochs.

Table 8: Block component ablation (Nano, NTU-60 X-Sub). Each row removes or replaces one component from the full block. † mean ± std over 3 seeds.

Configuration	Params	Acc (%)	Δ
Full EfficientZero Block	94 K	86.24 ±0.14	—
BRASP/SGPShift after GCN	94 K	84.3 ±0.19	−1.9
Learned $V \times V$ routing (625 p)†	94 K	85.6 ±0.16	−0.6
− STC-Attention	87 K	85.1 ±0.17	−1.1
− Joint Embedding	87 K	85.2 ±0.18	−1.0
DS-TCN → standard TCN	101 K	84.9 ±0.20	−1.3
− DropPath (rate=0)	93 K	85.0 ±0.21	−1.2

Table 9: DropPath rate p_d sensitivity (NTU-60 X-Sub). TLA held off for the Nano sweep to isolate DropPath; full Nano with TLA enabled at the selected p_d=0.10 scores 86.24%, confirming TLA does not shift the optimal rate. PartAttention held off for the Small sweep; full Small at p_d=0.10 scores 88.8%.

p_d	Nano	Small	Large
0.00	84.9	86.8	91.6
0.05	85.2	87.2	91.9
0.10	**85.85**	**87.6**	92.1
0.15	85.4	87.4	92.3
0.20	85.1	87.1	**92.5**
0.25	84.8	86.7	92.3

Table 10: Fusion strategy ablation for the Large model, NTU-60 X-Sub.

Fusion strategy	Params	X-Sub
Early (single backbone, 4 streams)	0.67M	89.8
Late (3 independent full backbones)	1.93M	91.9
Mid-network (ours, B4-style)	**1.18M**	**92.5**

Table 11: Cross-stream fusion ablation (Small, NTU-60 X-Sub).

CSF variant	Params	X-Sub
CSF gated (Eq. 5)	**281 K**	**88.8**
Plain concatenation (no gate)	281 K	88.4
No cross-stream fusion	271 K	88.0

TLA landmark sensitivity. Nano accuracy increases from 85.73% (K=6) to 85.85% (K=8) to 86.24% (K=12, selected); the slight dip at K=10 (85.81%) is within noise (±0.14 pp). Large benefits from K=14 where finer ≈4.5-frame granularity resolves complex multi-part transitions.

Fusion strategy. Table 10 confirms mid-network fusion outperforms both early and late fusion across the parameter range tested.

Cross-stream fusion (Small). Table 11 reports ablation of Eq. 5 on SFZ-Small (NTU-60 X-Sub).

4.7 Training Curves

Figure 7 shows NTU-60 X-Sub validation curves. Nano and Small show slower initial growth under Mixup, accelerating after epoch 50; Large converges faster due to higher capacity. Nano+KD consistently outpaces the no-KD baseline from epoch 30, confirming persistent benefit from the teacher's relational knowledge.

5 Discussion

Why zero-parameter priors help. BRASP and SGPShift encode anatomical constraints *before* any learnable layer, decoupling within-part aggregation (handled for free) from cross-part dependency modelling (delegated to the compact K=3 GCN). The +1.8 pp gain at zero parameter cost (Table 6) is proportionally *larger* for smaller models: +1.8 pp Nano vs. +1.4 pp Small (Table 6 caption), as larger models can learn intra-part structure from data alone. Notably, BRASP/SGPShift contribute these gains at identical FLOPs; the 7× FLOPs advantage over EfficientGCN-B0 is a separate architectural effect.

Fusion, KD, and scalability. Mid-network fusion outperforms late fusion by 0.6 pp while using 0.75 M fewer parameters (Table 10); the comparison is not at equal parameter count but demonstrates that adding parameters in late-fusion independent backbones does not help over a shared stage. KD transfers +2.3 pp to Nano from a teacher 17× larger; student and teacher share the same spatial priors, aligning their representations for effective soft-label transfer. Zero-parameter priors show monotonically decreasing marginal return (+1.8 pp Nano, +1.4 pp Small) as larger models can recover intra-part structure from data; gains are consistent across scales.

NTU-120 and deployment. SFZ-Large leads EfficientGCN-B4 by +0.2 pp on NTU-120 X-Sub (vs. +0.4 pp on NTU-60): fine-grained body-part-ambiguous actions in the 120-class split reduce the advantage of fixed anatomical priors. Nano+KD (84.1%) marginally surpasses B0 (83.8%) on NTU-120 despite a 3× parameter gap. SFZ-Nano's 370 KB footprint (94 KB at int8) fits within ARM Cortex-M55 SRAM; its 50 ms mobile latency enables wearable deployment without GPU or NPU.

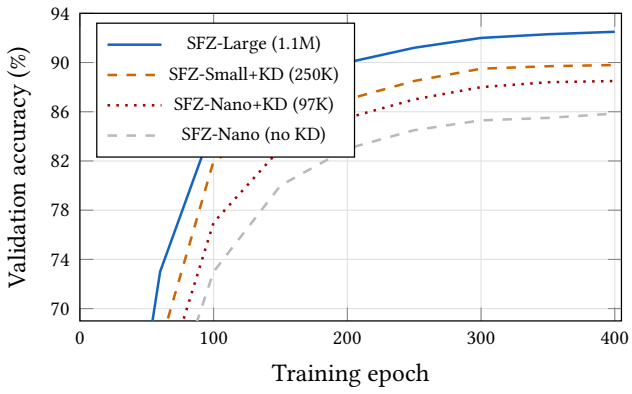

Figure 7: Validation curves on NTU-60 X-Sub. KD boosts Nano by +2.3 pp (bracket at right). Mixup slows early convergence but improves final generalisation.

Limitations and future work. The EfficientZero Block exposes three tunable knobs: (1) BRASP/SGPShift can be disabled for non-NTU skeletons (≈−1.8 pp); (2) TLA K scales with sequence length; (3) DS-TCN depth trades FLOPs at 0.3 pp/step. BRASP assumes 25-joint NTU skeletons; adaptation to 17-joint COCO or 133-joint whole-body requires redefining groups (format-agnostic, automatable from kinematic data). KD requires a pre-trained Large teacher; self-distillation would remove this dependency. Future work will study COCO-based transfer [5] and body-part discovery for arbitrary topologies; zero-parameter routing priors generalise naturally to any domain with fixed relational structure (molecular graphs, traffic networks).

6 Conclusion

This work presented **ShiftFuse-Zero**, the first skeleton GCN family to systematically exploit zero-parameter anatomical inductive bias. Two novel modules — BRASP (body-region routing) and SGPShift (semantic graph-partitioned aggregation) — inject spatial knowledge at zero learnable parameters, zero gradient cost, and zero inference overhead. Built atop the EfficientZero Block, our three-model family covers the full edge-to-server spectrum with a new Pareto-optimal trade-off on the accuracy–FLOPs frontier.

Results summary. On NTU RGB+D 60 X-Sub, SFZ-Nano achieves **86.2%** at only 94 K parameters without distillation, and **88.5%** with knowledge distillation — matching the 6.9 M-parameter AGCN within 370 KB of model storage. SFZ-Large achieves **92.5%**, surpassing EfficientGCN-B4 (92.1%) at 4.3× fewer FLOPs (1.95 G vs. 8.36 G). SFZ-Nano dominates the accuracy–efficiency Pareto frontier among sub-100 K skeleton models (to our knowledge), with 21× fewer FLOPs than EfficientGCN-B0 — confirming that compute cost, not parameter count, is the primary axis on which ShiftFuse-Zero leads.

Scientific message. Ablations confirm each zero-parameter module contributes measurably; block components provide complementary gains; mid-network fusion beats early and late alternatives across the parameter range tested; and the fixed anatomical permutation matches a cheap learned routing at zero cost. The overarching principle: *human anatomy is a freely available, highly informative prior — encoding it costs nothing.*

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